

Jorge Morais

FullStack & Embedded Systems Developer

Full-stack developer with 5+ years building high-performance web applications and embedded systems. From C-based payment machine APIs to React/Node.js platforms processing 900k+ documents. Specialized in performance optimization, real-time systems, and IoT protocols.

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PROFESSIONAL EXPERIENCE

Senior FullStack Developer

Jun 2023 - Present | Braga, Portugal

Balanças Marques

C • React • Node.js • TypeScript • MQTT • CCTalk • ID003 • Socket.IO • MySQL

- Architected **Alice Payment Machine API** in C with manual memory management (malloc/free), validated with Valgrind for zero memory leaks, implementing CCTalk/ID003 protocols for bill validators and control boards
- Engineered **Qore platform** memory footprint optimization, reducing consumption from **3GB to 850MB (72% reduction)** through efficient data structures and memory profiling
- Optimized mass data processing pipeline for **900k+ POS documents**, reducing execution time from **2 days to 2h (96% faster)** via concurrent processing and multithreading
- Built real-time IoT monitoring system for **50+ industrial scales** using MQTT and Socket.IO, and led code reviews establishing team development standards

Frontend Developer

Oct 2022 - May 2023 | Remote

Frontkom (Norwegian Tech Company)

(Madeira/Norway)

React • Vue.js • Drupal • TypeScript • Sass • Tailwind • Figma

- Developed **10+ enterprise websites** for Norwegian clients using React and Drupal headless CMS
- Implemented responsive, WCAG AA compliant components and optimized performance via code splitting and lazy loading
- Collaborated with international team across 3 time zones using Agile/Scrum methodologies

Web Developer

Aug 2021 - Oct 2022 | Guimarães, Portugal

Netgocio

Vue.js • JavaScript • PHP • MySQL • Bootstrap • Sass

- Managed **400+ hosting deployments** for B2B/B2C e-commerce platforms with **10k+ monthly users** and 99.9% uptime
- Optimized MySQL queries, reducing page load from **3.5s to 0.8s (77%)**, and built custom Vue.js components
- Implemented responsive design system using Sass and Bootstrap with mobile-first approach

EDUCATION & CERTIFICATIONS

CS50: Introduction to Computer Science (In Progress) | [Harvard University](#) | 2025

Front-end Professional Course - Level 4 EQF | [IwTraining](#) | 2019-2020

High School Diploma (12th Grade) | [Secondary Education](#)

Technical Certifications:

JavaScript • Git & GitHub • Data Structures & Algorithms • Web Design

TECHNICAL SKILLS

FRONTEND:

React • Vue.js • Next.js • TypeScript • JavaScript • HTML5/CSS3 • Tailwind • Sass

BACKEND:

Node.js • Express.js • REST APIs • GraphQL • Socket.IO • Microservices

EMBEDDED & IOT:

C • CCTalk • MQTT • Hardware Integration • Real-time Systems

DATABASE & DEVOPS:

PostgreSQL • MySQL • MongoDB • Docker • AWS • Git • CI/CD

LANGUAGES

Portuguese

Native / Bilingual

English

Professional Working

Spanish

Professional Working